TA Session

Question 1. Bottling plant and the sample mean

A beverage plant fills bottles with a long-run mean volume of 500 ml and population standard deviation of 12 ml. A quality inspector randomly selects 36 bottles. Assuming the CLT applies, find the probability that the sample mean is between 497 ml and 503 ml.

Detailed solution

Step 1: $\mu = 500$, $\sigma = 12$, $n = 36$.

Step 2: Standard error $SE = \sigma / \sqrt{n} = 12 / \sqrt{36} = 12 / 6 = 2$ ml.

Step 3: Lower Z-score $= (497 - 500) / 2 = -1.50$. Upper Z-score $= (503 - 500) / 2 = +1.50$.

Step 4: Z-table gives $\Phi(1.50) = 0.9332$. By symmetry, $\Phi(-1.50) = 1 - 0.9332 = 0.0668$.

Step 5: $P(497 \le \bar{x} \le 503) = 0.9332 - 0.0668 = 0.8664$.

Final answer: The probability is 0.8664, or 86.64%.

Question 2. Average exam score for a sample

At a large school, individual scores on a standardised test have population mean 68 marks and standard deviation 10 marks. A random sample of 25 students is selected. Using the CLT, estimate the probability that the sample mean score exceeds 71 marks.

Detailed solution

Step 1: $\mu = 68$, $\sigma = 10$, $n = 25$.

Step 2: $SE = 10 / \sqrt{25} = 2$ marks.

Step 3: $z = (71 - 68) / 2 = 1.50$.

Step 4: Z-table gives $\Phi(1.50) = 0.9332$, the area to the left of $z = 1.50$.

Step 5: $P(\bar{x} > 71) = 1 - 0.9332 = 0.0668$.

Final answer: The probability is 0.0668, or 6.68%.

Question 3. Delivery times and a sample average

A courier service has individual delivery times with mean 40 minutes and standard deviation 8 minutes. A supervisor takes a random sample of 64 deliveries. Find the probability that the sample mean delivery time is less than 38 minutes.

Detailed solution

Step 1: $\mu = 40$, $\sigma = 8$, $n = 64$.

Step 2: $SE = 8 / \sqrt{64} = 1$ minute.

Step 3: $z = (38 - 40) / 1 = -2.00$.

Step 4: Z-table gives $\Phi(2.00) = 0.9772$, so $\Phi(-2.00) = 1 - 0.9772 = 0.0228$.

Step 5: $P(\bar{x} < 38) = 0.0228$.

Final answer: The probability is 0.0228, or 2.28%.

Question 4. Weight of cereal packets

A machine produces cereal packets with population mean weight 750 g and standard deviation 15 g. A random sample of 100 packets is weighed. Assuming the CLT applies, calculate the probability that the sample mean lies between 748 g and 753 g.

Detailed solution

Step 1: $\mu = 750$, $\sigma = 15$, $n = 100$.

Step 2: $SE = 15 / \sqrt{100} = 1.5$ g.

Step 3: Lower limit $z_1 = (748 - 750) / 1.5 = -1.33$ (rounded). Upper limit $z_2 = (753 - 750) / 1.5 = 2.00$.

Step 4: Z-table gives $\Phi(2.00) = 0.9772$ and $\Phi(1.33) = 0.9082$. Thus $\Phi(-1.33) = 1 - 0.9082 = 0.0918$.

Step 5: $P(748 \le \bar{x} \le 753) = 0.9772 - 0.0918 = 0.8854$.

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Final answer: The probability is approximately 0.8854, or 88.54%, using z = −1.33 rounded to two decimal places.

Question 5. Sample mean of household electricity use

The daily electricity use of households in a town has mean 18 kWh and standard deviation 6 kWh. An analyst randomly samples 36 households. Using the CLT, find the probability that the sample mean exceeds 19.5 kWh.

Detailed solution

Step 1: $\mu = 18$, $\sigma = 6$, $n = 36$.

Step 2: SE = $6 / \sqrt{36} = 1$ kWh.

Step 3: $z = (19.5 - 18) / 1 = 1.50$.

Step 4: Z-table gives $\Phi(1.50) = 0.9332$.

Step 5: $P(\bar{x} > 19.5) = 1 - 0.9332 = 0.0668$.

Final answer: The probability is 0.0668, or 6.68%.

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Question 6. The 3-sigma range for a normal process

The diameter of a manufactured shaft is normally distributed with mean 20.0 mm and standard deviation 0.10 mm. Find the interval covering approximately 99.73% of shaft diameters using the 3-sigma rule. Then use the Z-table to show the probability inside the interval.

Detailed solution

Step 1: $\mu = 20.0$ mm and $\sigma = 0.10$ mm.

Step 2: Three-sigma limits = $\mu \pm 3\sigma = 20.0 \pm 3(0.10) = 20.0 \pm 0.30$, giving 19.70 mm to 20.30 mm.

Step 3: Standardised limits are $z = -3.00$ and $z = +3.00$.

Step 4: Z-table gives $\Phi(3.00) = 0.9987$. By symmetry, $\Phi(-3.00) = 1 - 0.9987 = 0.0013$.

Step 5: Using table values, $P(-3 \le Z \le 3) = 0.9987 - 0.0013 = 0.9974$. More precise normal probabilities give approximately 0.9973; the small difference is due to table rounding.

Final answer: The 3-sigma interval is 19.70 mm to 20.30 mm, with approximately 99.73% coverage.

Question 7. Probability of being outside three standard deviations

The fill volume from a calibrated process follows a normal distribution with mean 1,000 ml and standard deviation 4 ml. A bottle is flagged if its volume differs from the mean by more than three standard deviations. Estimate the probability that a randomly selected bottle is flagged, using the Z-table.

Detailed solution

Step 1: Three standard deviations = $3\sigma = 3(4) = 12$ ml.

Step 2: A bottle is flagged if $X < 988$ ml or $X > 1012$ ml.

Step 3: Standardised limits are $z = -3.00$ and $z = +3.00$.

Step 4: Z-table gives $\Phi(3.00) = 0.9987$. Upper tail = $1 - 0.9987 = 0.0013$; by symmetry, lower tail = 0.0013.

Step 5: Total outside probability = 0.0013 + 0.0013 = 0.0026 using rounded table values. The more precise 3-sigma rule gives about 0.0027 (0.27%).

Final answer: Using the table entries, the probability is approximately 0.0026 (0.26%); more precisely it is about 0.27%.

Question 8. Six-sigma limits for a precision component

A precision component has a normally distributed length with mean 50 mm and standard deviation 0.02 mm. The quality team defines its six-sigma limits as six standard deviations on either side of the mean. Calculate the limits and estimate the probability that a component lies within them. Explain the limitation of a standard Z-table.

Detailed solution

Step 1: $\mu = 50$ mm and $\sigma = 0.02$ mm.

Step 2: Six-sigma distance = $6\sigma = 6(0.02) = 0.12$ mm.

Step 3: Lower limit = $50 - 0.12 = 49.88$ mm; upper limit = $50 + 0.12 = 50.12$ mm.

Step 4: Standardised limits are $z = -6.00$ and $z = +6.00$.

Step 5: Most classroom Z-tables stop around $z = 3.99$ or 4.00. To four decimal places, $\Phi(6.00) = 1.0000$. The upper-tail probability is approximately 9.9×10^{-10}; the lower tail is the same.

Step 6: $P(-6 < Z < 6) \approx 1 - 2(9.9 \times 10^{-10}) = 0.9999999980$, or about 99.9999998%. A four-decimal Z-table simply rounds this to 1.0000.

Final answer: The six-sigma limits are 49.88 mm to 50.12 mm. The coverage is approximately 99.9999998%; a conventional four-decimal table shows 1.0000.

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Question 9. Comparing 3-sigma and 6-sigma defect rates

A normally distributed process has mean 200 units and standard deviation 5 units. A product is outside specification if its value is below $\mu - 3\sigma$ or above $\mu + 3\sigma$. Find the probability of a product falling outside specification using the Z-table. Then compare this with the outside probability for a centred six-sigma interval.

Detailed solution

Step 1: Three-sigma limits = 200 ± 3(5) = 185 and 215 units.

Step 2: The corresponding Z-scores are −3.00 and +3.00.

Step 3: Z-table gives $\Phi(3.00) = 0.9987$, so $P(Z > 3) = 1 - 0.9987 = 0.0013$. By symmetry, $P(Z < -3) = 0.0013$.

Step 4: Total outside $\pm 3\sigma = 0.0013 + 0.0013 = 0.0026$ from rounded table values, or about 0.27% more precisely.

Step 5: For a centred $\pm 6\sigma$ interval, each tail is approximately 9.9×10^{-10}, so total outside probability is approximately 1.98×10^{-9} under an ideal normal model.

Final answer: Outside $\pm 3\sigma$, probability is about 0.0026 from the rounded table (about 0.27% more precisely). Outside $\pm 6\sigma$, it is approximately 1.98×10^{-9}.

Question 10. CLT and three standard errors

The population of transaction amounts at a retail store has mean ₹800 and standard deviation ₹120. A random sample of 64 transactions is selected. Using the CLT, find the probability that the sample mean is within three standard errors of the population mean. State the Z-table entries and explain how this relates to the 3-sigma rule.

Detailed solution

Step 1: μ = ₹800, σ = ₹120, n = 64.

Step 2: Standard error = $\sigma / \sqrt{n} = 120 / 8$ = ₹15.

Step 3: Three standard errors = 3(15) = ₹45. The interval for the sample mean is ₹800 ± ₹45, i.e. ₹755 to ₹845.

Step 4: z for ₹755 = (755 − 800) / 15 = −3.00; z for ₹845 = (845 − 800) / 15 = +3.00.

Step 5: Z-table gives $\Phi(3.00) = 0.9987$ and $\Phi(-3.00) = 0.0013$.

Step 6: Probability = 0.9987 − 0.0013 = 0.9974 from rounded table values; more precise 3-sigma coverage is about 0.9973.

Final answer: The probability is approximately 0.9973 (99.73%); subtracting the rounded table entries gives 0.9974.

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Z-TABLE WORKING SUMMARY

For a cumulative standard normal table $\Phi(z) = P(Z \le z)$:

- Left-tail probability: $P(Z < z) = \Phi(z)$.
- Right-tail probability: $P(Z > z) = 1 - \Phi(z)$.
- Probability between two Z-scores: $P(a < Z < b) = \Phi(b) - \Phi(a)$.
- Symmetry: $\Phi(-z) = 1 - \Phi(z)$.
- Central interval: $P(-z < Z < z) = 2\Phi(z) - 1$.

Example: $z = 1.50$ is found at row 1.5 and column 0.00; the cumulative table entry is $\Phi(1.50) = 0.9332$.

Important note: The centred $\pm 6\sigma$ interval gives approximately 99.9999998% coverage for an ideal normal distribution. Some industrial Six Sigma calculations assume a 1.5σ process shift; that convention is different from the centred $\pm 6\sigma$ calculation used here.

Exam tip: Write the Z-score, state the exact Z-table value, identify whether the required probability is a left tail, right tail, or interval, and then perform the subtraction or addition.

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